MILITARY SPECIFICATION

# DIGITIZER, VOICE: FULL AND HALF DUPLEX SPEECH COMPRESSION AND DIGITAL ENCODING

## 1. SCOPE

1.1 This specification describes the performance, interface, and test requirements for a voice digitizer (VODIG). The VODIG equipment shall provide a full and half duplex speech compression and digital encoding capability at 2400 bits per sec. VODIG will be employed as part of a secure voice system in both HF and wireline terminals.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the data of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

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| MIL-S-901 | - Shock Tests, H.I. (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for |
| MIL-Q-9858 | - Quality Program Requirements |
| MIL-E-16400 | - Electronic Equipment, Naval Ship and Shore: General Specification |
| MIL-E-17555 | - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of |
| MIL-M-38793 | - Manuals, Technical; Calibration Procedures, Preparation of |
| MIL-M-38793/1 | - Manuals, Technical; Calibration Procedures, Submission |

STANDARDS

MILITARY

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| MIL-STD-12 | - Abbreviations for Use on Drawings and in Technical Type Publications |
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-108 | - Definition of and Basic Requirement for Enclosure for Electrical and Electronic Equipment |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-188 | - Military Communication System Technical Standards |
| MIL-STD-189 | - Racks, Electrical Equipment, 19 inch and Associated Panels |
| MIL-STD-198 | - Capacitors, Selection and Use of |

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MIL-STD-199 - Resistors, Selection and Use of
MIL-STD-242 - Electronic Equipment Parts, Selected Standards
MIL-STD-415 - Test Provisions for Electronic System and Associated Equipment, Design Criteria for
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics Measurement of
MIL-STD-470 - Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471 - Maintainability Demonstration
MIL-STD-701 - List of Standard Semiconductor Devices
MIL-STD-781 - Reliability Test, Exponential Distribution
MIL-STD-785 - Reliability Program for System and Equipment Development and Production
MIL-STD-794 - Part and Equipment, Procedures for Packaging and Packing of
MIL-STD-810 - Environmental Test Methods
MIL-STD-1345 - Data Measurement in Support of Maintenance, Calibration and Repair of Electronic Equipment
MIL-STD-1364 - Standard General Purpose Electronic Test Equipment
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment
MIL-HDBK-472 - Maintainability Prediction

PUBLICATIONS

NAVPERS 18068 - Manual of Qualification for Advancement in Rating
NACESM 5100 - Electromagnetic, Compromising Emanations Laboratory Test Standard

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.



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UNITED STATES OF AMERICA STANDARDS INSTITUTE
ANSI Y32.16-1968 - Electrical and Electronics
Reference Designations.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 General requirements. The equipment shall meet the requirements of MIL-E-16400, to the extent specified herein.

3.1.1 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.3).

3.1.2 General description. The VODIG equipment covered by this specification shall be a rugged, lightweight, versatile, micro-miniaturized design. The VODIG will be used to provide digitized voice at 2400 bits per sec (bps).

3.1.3 System compatibility. The equipment shall be compatible with KG-30 series electronic key generators and digital data modems MD-923/U and AN/ACQ-5.

3.1.4 Contractor responsibility. All hardware, cable, connectors, parts, material and electronics required to make the VODIG a complete and operable unit as specified herein shall be provided by the contractor. In addition, it shall be the responsibility of the contractor to provide all cables, connectors, test instrumentation and test facilities and other devices required to set up and perform all tests at the contractor's plant or other place of test, to show compliance with the requirements specified herein.

3.1.5 Implementation. Implementation of the VODIG may either be analog, digital or a combination. However, the use of a central processing unit (CPU) to perform the voice digitized function is prohibited.

3.2 Performance requirements.

3.2.1 Modes of operation. The equipment shall be capable of operating either full or half duplex. In addition, a back-to-back internal loop test shall be provided for test purposes with the sidetone muted.

3.2.2 Intelligibility. The intelligibility of the VODIG shall be at least 75 percent as measured by the Diagnostic Rhyme Test at 2400 bps back-to-back, and at least 70 percent at $10^{-2}$ bit error rate.

3.2.3 Vogad. An automatic (voice operated) gain control device (Vogad) shall be provided to compensate for variations in average speech levels. The operating range of the Vogad shall be -31.5 dBm to -8.5 dBm.

3.2.4 Pitch-extractor. The pitch extractor shall automatically and continuously track the fundamental frequency of the speaker's voice over a range of at least 70 to 300 Hertz (Hz).



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3.2.5 Adjustments. There shall be no adjustments in the VODIG other than input and output level and the test, measurement, and diagnostic equipment (TMDE).

3.2.6 Handset. Operator audio interface to the VODIG shall be by means of a contractor furnished handset. This handset shall include a microphone, earphone, and a momentary push-to-talk switch and shall be a Western Electric G1 or equivalent. Connection of the handset to the VODIG shall be by a coiled cable, approximately one foot long when retracted and at least four feet long at fullest extension. When not in use, the handset shall rest in a hanger on the front panel.

3.2.7 Sidetone. The level shall be nominally from minus 3 decibels (db) to minus 6 db relative to the received signal.

3.2.8 Loudspeaker. A loudspeaker shall be provided on the front panel in order to listen to the synthesized voice messages. A volume control shall be provided on the front panel to control speaker volume from off through maximum volume. The maximum intelligible volume shall be at least 76 db above $10^{-16}$ watt/cm$^2$ measured at a distance of 2 meters in a sound absorbing booth or chamber when the volume control is at the maximum setting.

3.2.9 Data, control, and timing interface. The VODIG shall provide the standard low level (Polar) interfaces for all data, control, and timing in accordance with MIL-STD-188.

3.2.10 Clock and data phase relationship. Clock and data phase relationship shall be in accordance with MIL-STD-188. In addition, the opposite phase for clock shall be strap selectable.

3.2.11 Back-to-back timing requirements. An internal frequency source shall provide the clock required for digital back-to-back operation of a single equipment.

3.2.12 Transmitter timing. The analyzer (transmitter) shall operate from an externally provided 2400 baud MIL-STD-188 low level timing signal.

3.2.13 Receiver timing. The synthesizer (receiver) shall operate from an externally provided MIL-STD-188 low level signal at 2400 baud.

3.2.14 Synchronization. The VODIG shall sync in less than 8 frames containing frame bits only with no errors present. The VODIG shall sync in less than 300 msec, 95 percent of the time when the bit error rate is 10 percent or less.

3.2.15 Pre-synchronization tone. Upon receipt of data consisting of only binary ones, the Vodig shall apply a steady audio tone to the audio outputs. This tone shall cease when voice information is detected in the data.

3.2.16 Out of synchronization detection. An out of synchronization detector shall be provided to indicate when the VODIG is out of synchronization. A front panel indicator and signal to an output terminal shall be provided to indicate when the detector is activated.



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3.2.17 Push-to-talk (PTT). The VODIG shall provide a PTT control signal which is active when the local or remote PTT is activated and complies with MIL-STD-188.

3.2.18 Transmit frames only. The VODIG shall inhibit all data bits in the PCM stream and only transmit synchronization bits when the transmit frames-only control line is activated.

3.2.19 Synthesizer inhibit. The VODIG shall inhibit the synthesizer output when the synthesizer inhibit control line is activated.

3.2.20 Remote audio output. A remote audio output shall be provided. The output level shall be 0 dbm nominal and active at all times. The output impedance shall be 600 ohm ±10 percent and shall be balanced to ground.

3.2.21 Remote audio input. A remote audio input to the analyzer shall be provided and shall be activated at all times. The input level shall be 0 dbm peak and the input impedance shall be 600 ohm ±10 percent and shall be balanced to ground.

3.2.22 Remote PTT. A remote PTT input shall be provided. The input from the remote source shall be a contact closure when active and an open circuit (10,000 ohm) when inactive. The voltage across the contacts shall not exceed 30 volts.

3.2.23 Control tone input. An input shall be provided to allow direct application of an externally produced audio tone to all the Vodig audio outputs. The input shall accept signals of 0 dBm, and shall have an input impedance of 600 ohms, balanced to ground.

3.3 Electrical design.

3.3.1 The equipment shall be designed to operate with input power of nominal 115 ±10 percent alternating current (VAC), 50 Hz - 5 percent to 400 Hz + 10 percent single-phase.

3.3.2 The equipment shall be designed to operate within of the Primary power supply line voltage and frequency requirements of MIL-E-16400.

3.3.3 Primary power circuits. The equipment shall be designed to meet the primary power circuits requirements of MIL-E-16400.

3.3.4 Overload protection. Overload protection shall be in accordance with the requirements of MIL-E-16400. Overload indication shall be provided.

3.4 Reliability.

3.4.1 Quantitative reliability requirement. The specified MTBF ($\theta o$ as defined by MIL-STD-781) of this equipment shall be 4000 hours.

3.4.2 Reliability program plan. The contractor shall prepare a reliability program plan in accordance with and in paragraph sequence with



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MIL-STD-785. Particular attention should be paid to those sections of MIL-STD-785 which require submission of detailed tasks, procedures to implement and control these tasks, and time schedules. The reliability program plan shall be prepared as a separate and complete entity within the total system or equipment project delineation. The plan, including reliability test plans, must describe an integrated effort within the total project; it shall provide information to show how the contractor will meet specified quantitative reliability requirements during development and manufacture.

3.4.3 Reliability prediction. The contractor shall prepare a reliability prediction using the method and K factor for shipboard equipment of MIL-HDBK-217 entitled, *How to Make a Reliability Prediction Based on Stress* Factors and Part Populations. The prediction shall assume a 50°C ambient temperature plus the estimated internal heat rise of the equipment. The contractor shall evaluate the prediction with respect to the reliability expected from a production unit. If there is a discrepancy between the predicted MTBF and the specified MTBF the contractor shall identify the causes of the discrepancies and the proposed corrective action required to ensure that the specified MTBF of the equipment is attained. The processes and procedures required to ensure production of reliable equipment shall be thoroughly documented. (see 6.2).

3.4.4 Source of failure rates. The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned. When such alternate sources of data are proposed, enough information shall be furnished to enable the command or agency concerned to evaluate the validity and applicability of the results derived therefrom.

3.4.5 Reliability reports. Reports providing a complete accounting of progress on each element defined by the reliability program plan, results achieved, and status of actions to resolve problems and correct weak links shall be prepared (see 6.2).

3.5 Maintainability.

3.5.1 Maintenance concept. The maintenance concept for this equipment is for the removal and replacement of plug in subassemblies (printed circuit cards) or modules for restoring equipment to specification operating requirements. The equipment maintenance shall be facilitated by providing self-test features which permit isolation of failures to the printed circuit card or module level. Built-in self-test features shall provide indication of operational readiness when repairs have been completed. Fault localization and isolation (trouble shooting) procedures for replacement of plug-in subassemblies or modules should not exceed skills or capabilities of a Navy Oceanographic Technician Third Class (OT-3). Skills and capabilities of Navy Oceanographic Technicians are as outlined in NAVPERS Manual 18068. Module and printed circuit card repairs will be accomplished by the intermediate maintenance service provided at the installation site or at the depot depending upon the degree of rework required.

3.5.2 Quantitative maintainability requirement. The equipment shall have a specified equipment repair time (ERT) of 15 minutes with repair accomplished



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by the replacement of modules, chassis-mounted electronic, electromechanical, and mechanical parts. The specified ERT for the repair of each module shall be 120 minutes. The ERT includes localization, isolation, disassembly, interchange, reassembly, and checkout for all maintenance tasks.

3.5.3 Qualitative maintainability requirements. The equipment shall be constructed to provide ease of maintenance, accessibility and replacement of all modules and parts.

3.5.4 Maintainability program plan. The contractor shall prepare a maintainability program plan in accordance with the requirements of MIL-STD-470. The maintainability program plan shall describe the tasks to be performed and the procedures for conducting and controlling the maintainability program. The plan shall address each paragraph of MIL-STD-470 in sequence. The plan shall define and identify the interface between the maintainability and integrated logistic support requirements (See 6.2).

3.5.4.1 Maintainability prediction and analysis.

3.5.4.1.1 Maintainability prediction. A maintainability prediction (see 6.2) shall be performed using Part A (to determine the $MTTR_G$) and B (to determine the mean preventive maintenance) Procedure II of MIL-HDBK-472. The contractor shall document and justify all assumptions and the applicability of all data used in the prediction. The prediction shall include all mechanical, electromechanical, and electronic parts. The prediction shall be prepared in two parts. Part I (equipment repair) shall be performed for replacement of modules and chassis-mounted components and parts at the organizational level of maintenance. Part II (module repair) shall be accomplished by the replacement of piece parts at the organizational, intermediate or depot-level of maintenance as appropriate. The results of the prediction shall be compared to the required ERT. Where the predicted values do not achieve the requirements the contractor shall accomplish such changes in design as are necessary to improve the predicted values sufficiently to achieve the requirements.

3.5.4.1.2 Maintainability analysis. A maintainability analysis (see 6.2) shall be performed concurrently with the design effort for the incorporation of the quantitative and qualitative maintainability requirements into the equipment design. The maintainability analysis shall evaluate equipment design and design changes along with the possible catastrophic and critical modes of equipment failure. The maintainability analysis shall also include the following:

(a) Task and skill analysis for all repair actions required to be performed by service technicians.
(b) A study of the indications and indicators of failure present at the operator level and at the various levels of maintenance.
(c) A listing of special tools and test equipment (including special alignment jigs and fixtures) required during maintenance of the equipment.
(d) Review of potential design, maintenance of production problems.
(e) Identification of the principal items inhibiting maintainability achievements and proposed solutions.
(f) Corrective and preventive maintenance features.



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3.5.4.2 Maintainability demonstration tests. The maintainability requirements shall be demonstrated in accordance with 4.8.

3.5.4.3 Maintainability reports. Reports in accordance with MIL-STD-470 that provide a complete accounting of progress on each element of the plan, results achieved, procedures used, and status of actions to resolve problems and correct deficiencies shall be prepared (see 6.2).

3.5.5 Construction. The units comprising a complete equipment shall be designed for rack mounting in accordance with MIL-STD-189.

3.5.5.1 Dimensions and weight. The maximum dimensions and weight of the receiving equipment, including the cabinet, shall be as follows:

(a) The depth of the VODIG equipment shall not exceed 24 inches (60.96 cm) including mating connectors and cable egress allowance.
(b) The height of the VODIG equipment shall not exceed 7.00 inches (17.78 cm)
(c) The weight of the VODIG shall not exceed a 60 pound (27.216 kg) maximum limit.

3.5.5.1.1 Chassis. Each chassis shall be designed such that it can be completely serviced from the front of the equipment. Complete slide rail mechanisms shall be provided to permit withdrawal of each chassis for servicing. A cable retracting mechanism shall be provided to guide cables when the equipment is withdrawn for servicing. Chassis tilting and locking facilities shall be provided for access to the underside of the chassis assembly for inspection or maintenance. Automatically operated locks shall be provided to lock the chassis in the servicing position.

3.5.5.1.2 Rear panel. The rear panel shall contain connectors for the digital signal and control, analog signal and control, the remote functions and the power. The connectors shall be of the MS-3106 series type. The equipment shall be supplied with all mating connectors.

3.5.1.1.3 Electrical interconnections. The electrical interconnections between functional modular assemblies shall be made by means of multicircuit connectors to permit replacement of assemblies without the necessity for soldering of connections.

3.5.5.2 Printed wiring boards. All active components, with the exception of power supply components, shall be mounted on plug-in printed circuit (PC) boards. The number of PC boards shall not exceed 16. For maintainability demonstration purposes, the PC boards may be excluded from the field reparable category, but considered reparable at higher levels.

3.5.5.2.1 PC board interchangeability. Each PC board in the VODIG shall be interchangeable with any other like boards. Board replacement in multiples, such as matched sets, is not permissible.



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3.5.5.2.2 PC boards. All PC boards shall be of the double or single side copper type. Multilayer boards shall not be used. Design shall be such that packaging to an ATR configuration is possible.

3.5.5.2.3 PC card locks. Suitable means to securely lock the PC cards in their sockets shall be provided.

3.5.5.2.4 Extender boards and cables. Extender boards and cables shall be provided with each voice digitizer to extend all modules and PC boards if necessary for servicing. Provisions shall be made for storing of extender boards and cables inside the cabinet.

3.5.5.2.5 PC board extraction. All PC boards shall be removable without the use of any tools or extractors except as may be a part of the board itself.

3.5.5.3 Controls, indicators and panel layouts. The design of operator and maintainer panel layouts and controls shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the control, indicator and panel layout requirements of MIL-E-16400 except that the color coding shall be in accordance with MIL-STD-1472. Panel abbreviations shall be in accordance with MIL-STD-12.

3.5.5.3.1 Specific design requirements. The principal human interaction between the operator and the VODIG unit consists of the selection of the unit operational modes and the use of the voice digitizer hand microphone. Both the unit mode selection controls and the voice digitizer operator hand units shall be easily accessible by an operator. The maintenance and self-test controls are excluded from this requirement.

3.5.5.3.2 Front panel controls and indicators. The front panel shall include the following:

(a) Power On/Off control
(b) Power On/Off indicator
(c) Overload indicator
(d) Speaker volume control
(e) Loudspeaker
(f) Sync indicator
(g) Operate/back-to-back/fault check selector switch
(h) Fault indicator

3.5.5.4 Panel mounted devices. Panel mounted potentiometers, switches (including toggle switches), and other similar devices shall be mechanically keyed to preclude the loosening of their respective securing devices during normal usage.

3.5.6 Parts. Parts used in the equipment shall be selected in accordance with the specified performance requirements and the meantime between failure (MTBF) of the equipment. Maximum use shall be made of parts selected from MIL-STD-242 and all parts shall have been selected to insure that adequate mechanical and electrical ratings are provided conforming with the parts



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requirements of MIL-E-16400. All integrated circuits and semiconductors shall be available from two or more independent sources with independent manufacturing facilities. Exceptions shall be approved by the command or agency concerned.

3.5.6.1 Non-standard parts. A listing of all non-standard parts shall be prepared for approval by the command or agency concerned (see 6.2).

3.5.6.1.1 The performance of the equipment shall not be dependent on the selection of individual resistors, transistors, capacitors, or other parts.

3.5.6.1.2 For like equipments, all essential parts, including removable units, repair parts and other items shall be interchangeable physically and electrically with corresponding items or parts without drilling or filing, or the use of undue force. In the design of the equipment, provisions shall be made for design tolerances sufficient to accommodate various articles such as transistors, resistors, capacitors and other parts involved having listed dimensions and without departure from the specified performance.

3.5.6.1.3 In the selection of parts, the widest tolerances possible commensurate with the particular long term circuit stability requirement shall be used. However, in the fabrication of the equipment, it is permissible to substitute parts of closer tolerances as long as interchangeability is not affected.

3.5.6.2 In order to facilitate maintenance, each part when assembled in a unit and set shall be identified by an appropriate reference designation in accordance with ANSI Y32.16-1968.

3.5.6.3 The arrangement of parts shall be such that replacement or adjustment of any part is possible without removal of or damage to adjacent parts. All parts shall be readily accessible for replacement or repair. Access to all parts, with chassis withdrawn or access doors open, shall be from the front of the cabinet.

3.5.6.4 Electron tubes. All active circuits, (rectifiers, amplifiers, oscillators, and so forth) shall employ solid state devices (transistors and diodes). Thermionic electron tubes will be permitted only if the equipment designer can demonstrate that their use will provide performance superior to the performance which can be obtained with transistors. Thermionic electron tubes shall not be used without the approval of the command or agency concerned (see 6.2).

3.5.6.5 Semiconductor devices. Semiconductor devices (diodes and transistors) shall be selected from the approved types in accordance with MIL-STD-701.

3.5.6.6 Transistors. Transistors shall be secured in approved sockets or chassis mounted heat sinks. Soldering of transistor leads to terminals will not be permitted for transistor applications other than those which are mounted in, and constitute parts of, replaceable plug-in cards or encapsulated modular assemblies.



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3.5.6.7 Microelectronic devices. Microelectronic devices selected shall be approved by the command or agency concerned (see 6.2).

3.5.6.8 Encapsulation. Plastic encapsulated semiconductor devices shall not be used.

3.5.6.9 Electrolytic capacitors. The use of aluminum foil electrolytic capacitors of any description will not be permitted in any circuitry not directly identified with the primary power supply. Tantalum capacitors may be employed to the extent permitted by MIL-E-16400.

3.5.6.10 Capacitors. Capacitors shall conform to the requirements of MIL-STD-198.

3.5.6.11 Resistors. Resistors shall conform to the requirements of MIL-STD-199.

3.5.6.12 Workmanship. Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

3.5.6.13 Soldering. Soldering shall be in accordance with Requirement 5 of MIL-STD-454.

3.5.6.14 Environmental conditions. The equipment shall operate in accordance with the performance requirements specified herein under the following ambient conditions.

3.5.6.14.1 Operating temperature. The equipment shall provide reliable operation in any ambient temperature between 0° and +50° C.

3.5.6.14.2 Storage temperature. The equipment shall not be damaged nor shall the operational performance be degraded when restored to the operating temperature range after being exposed for long periods in the non-operating range of -50° C to +75° C.

3.5.6.14.3 Humidity. The equipment shall meet the requirements of MIL-E-16400.

3.5.6.14.4 Shock. The equipment shall meet the shock requirements of MIL-E-16400 with the following exception: the 5-foot hammer blow test of MIL-S-901 shall be deleted. Analytical demonstration shall not be acceptable.

3.5.6.14.5 Vibration. The equipment shall meet the vibration requirements of MIL-E-16400.

3.5.6.14.6 Duty cycle. For normal operation the duty cycle shall be continuous.

3.6 Tempest and EMC.

3.6.1 Tempest. The VODIG shall be designed such that the level of Compromising Emanations (CE) generated shall be limited to the levels referenced in 8.1.1, 8.2, 8.3 and 8.4 of NACSEM 5100.

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3.6.2 Electromagnetic compatibility (EMC). The equipment shall meet the limit requirements for Class IC equipment in accordance with MIL-STD-461 for the following tests only:

CE01, CE02, CE03, CE04, CS01, CS02, CS06, RE01, RE02, RS01, RS02. and RS03

The equipment shall be self-compatible and compatible with co-located equipment normally working in its intended installation.

3.7 Safety. Safety shall be in accordance with MIL-E-16400.

3.7.1 Ground terminal. An electrical ground terminal shall be provided on the rear of the equipment for connecting the chassis to an external ground buss.

3.8 Test, measurement, and diagnostic equipment (TMDE).

3.8.1 Circuit test point design criteria.

3.8.1.1 Self-test. A front panel control and indicator shall be provided to permit the operator to determine if the VODIG is properly performing with the probability of indicating a fault when a fault has occured of 95 percent and the probability of indicating a fault when none has occurred of .01 percent. This test shall be automatic and shall be in addition to the back-to-back test.

3.8.1.2 Circuit test point accesses. Circuit test point accesses shall be provided which comply with the criteria outlined in MIL-STD-415. Test point data shall be prepared in accordance with MIL-STD-1345. Specific identifiable test points or test jacks or both, shall be provided for the following tests:

(a) System Circuit Performance
(b) LRI Fault Location
(c) Shop Line Replaceable Item Quality and Performance Tests
(d) Shop Replaceable Item and Fault Isolation.

3.8.2 Calibration.

3.8.2.1 Calibration and test procedures. The contractor shall develop, and prepare calibration test procedures required to confirm accuracy of built-in TMDE, portable TMDE and other measurement devices designed into the prime system equipment circuits. (see 6.2).

3.8.2.2 Instrument calibration procedures (ICP). The command or agency concerned shall review the calibration test procedures and based on the test equipment and measurements techniques utilized, shall indicate the TMDE for which ICPs are required. ICPs shall be developed, and prepared in accordance with MIL-M-38793 and MIL-M-38793/1A when required by the contract or order. (see 6.2). Calibration test procedures which do not require a separate ICP shall be included in the prime system equipment technical manual.

3.8.2.3 Calibration design criteria. All built-in TMDE shall be provided with either test points or test jacks or both on the front panel to isolate the TMDE from the prime circuit and to inject test signals into the TMDE for purposes



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of calibration, unless specifically waived by the command or agency concerned. All built-in TMDE shall be constructed to permit removal from the prime system equipment for calibration without affecting the operation of the system, where feasible. The built-in TMDE shall not be affected by the prime system operational environment during alignment or adjustments. All TMDE alignment and adjustment features shall be organized to enable the technician to view all pertinent instrumentation at the same time he is making TMDE adjustments.

3.8.3 LRI fault location.

3.8.3.1 LRI fault location test procedures. The contractor shall develop, and prepare the test procedures required to isolate the fault to a particular LRI. (see 6.2).

3.8.3.2 Design criteria for LRI fault localization. Either test points or test jacks or both, shall be provided to permit the measurement of signals at the LRI input and output terminals to demonstrate that the LRI does or does not have a failure.

3.8.4 Shop LRI performance and quality.

3.8.4.1 Shop LRI performance and quality test procedures. The contractor shall develop, and prepare test procedures required to demonstrate that an LRI which has been repaired or restored at the depot level is operating properly when the LRI is separated from the prime equipment. (see 6.2).

3.8.4.2 Shop LRI performance and quality design criteria. Either test points or test jacks or both, for shop LRI and submodule boards performance and quality test procedures shall be provided as a part of the connector of the LRI which will permit each LRI and submodule to be tested with either programmable or manual TMDE.

3.8.5 Shop replaceable item fault isolation and test procedures. The contractor shall develop and prepare all test procedures for shop replaceable item isolation required to isolate a throw-away part that has failed in an LRI or system. (see 6.2).

3.8.6 Special tools and test equipment. The following special tools and test equipment shall be supplied by the contractor:

(a) A complete set of extender boards, cables, and special purpose electronic test equipment required at the organization level of maintenance shall be supplied for each equipment. The command or agency concerned will indicate the activities to which the extender board, cables, and special purpose electronic test equipment sets are to be delivered. Special purpose electronic test equipment is defined as all test equipment not listed in MIL-STD-1364, manufactured by the contractor and solely for the support of the system and the equipment.

(b) A complete set of special tools shall be supplied with each equipment. Special tools are defined as those tools not listed in the federal supply catalog (copies of the catalog may be

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consulted in the Office of the Defense Contract Administration Service (DCAS)). The command or agency concerned will indicate the activities to which the special tool sets are to be delivered.

(c) All jigs, fixtures, special purpose test equipment and other interface devices developed by the contractor for testing, inspection and checkout of the equipment, shall be delivered to the command or agency concerned at the termination of the contract.

3.8.6.1 Selection of general purpose electronic test equipment (GPETE). GPETE shall be selected from MIL-STD-1364. Authorization to use GPETE not listed in MIL-STD-1364 for support testing, shall be requested in writing in accordance with Figures 1 and 7 of MIL-STD-1345.

3.8.7 Technical manual test procedures.

(a) Test procedures developed for circuit performance, calibration, and line replaceable item fault location shall be incorporated into the equipment technical manual.

(b) Test procedures developed for shop replaceable item fault isolation shall be incorporated into the equipment repair manuals.



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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality program requirement. The contractor shall provide and maintain a written quality program acceptable to the Government for supplies and services covered herein. The quality program shall be in accordance with MIL-Q-9858.

4.1.2 Quality plan. A quality plan, applicable to the equipment covered herein, shall be prepared for Government approval. (see 6.2)

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection system are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Test procedures. The contractor shall prepare test procedures for Government approval covering first article and quality conformance inspections. (see 6.2)

4.1.5 Quality assurance terms and definitions. Quality assurance terms and definitions used herein shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The method of examination and testing shall be classified as follows:

(a) First article inspection
(b) Quality conformance inspection

(1) Production inspection
(2) Production control inspection
(3) Environmental inspection
(4) Inspection of preparation for delivery.

(c) Reliability testing



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4.3 Unless otherwise specified (see 6.1), one unit shall be required for first article inspection. First article inspection consists of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in Table I.

Table I - First Article Inspection

| Inspection | Reference |
|---|---|
| Test jacks | 3.8.1.2 |
| Production inspection | 4.4.1 |
| Production control inspection | 4.4.2 |
| Environmental | 4.4.3 |
| Reliability tests | 4.5 |
| EMC | 4.4.2.3 |
| Tempest | 4.4.2.4 |
| Maintenance tests | 4.8 |

4.3.1 First article inspection report. At the conclusion of the first article inspection the contractor shall prepare a final report. The report shall contain test results, analysis, conclusions and recommendations. For convenience, the report may be divided into sections covering major test segments such as performance testing, environmental testing, electromagnetic testing, and tempest testing (see 6.2).

4.3.2 First article approval. Approval of the first article shall be by the command or agency concerned upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the first article. Prefabrication of production equipment prior to the Governments approval of the first article is at the contractors own risk. Following Government performed service approval tests the first article shall be returned to the contractor for his use in the fabrication and test of equipment submitted for acceptance. The first article may be reworked by the contractor and submitted for acceptance as a production equipment.

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every production equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the ommissions and errors of the production process. The inspection shall include functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests and examinations approved in accordance with 4.1.4 and shall include the requirements of Table II.



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Table II - Production Inspection

| Inspection | Reference |
|---|---|
| Visual Mechanical | |
| Workmanship | 3.5.6.12 |
| Soldering | 3.5.6.13 |
| Safety | 3.7 |
| Ground terminal | 3.7.1 |
| Front panel controls | 3.5.5.3.2 |
| Synchronization (Synch) | 3.2.15, 3.2.14 |
| Out of synch detection | 3.2.16 |
| Self test | 3.8.1.1 |

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to Table III and to the inspection procedures of MIL-STD-105, using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be approved in accordance with 4.1.4. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

Table III - Production Control Inspection

| Inspection | Reference | AQL% |
|---|---|---|
| Dimensions and weight | 3.5.5 | 6.5 |
| Supply line voltage and frequency | 3.3.2 | 6.5 |
| Back-to-back operation | 3.2.1, 3.2.11 | 6.5 |
| Vogad operation | 3.2.3 | 6.5 |
| Data control and timing interface | 3.2.9, 3.2.10 | 6.5 |
| Loudspeaker | 3.2.8, 4.4.2.6 | 6.5 |
| Input power | 3.3.1 | 6.5 |

4.4.2.1 Production control inspection report. At the conclusion of each production control inspection the contractor shall prepare a test report. The report shall include test results versus requirements and shall be correlated to the pertinent production inspection report (see 6.2).

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to the original specification requirements to correct the defective units and reinspect each equipment so reworked. If, on re-inspection of the rework, the equipment is acceptable, it may be provided on the contract as new equipment.

4.4.2.3 Electromagnetic compatibility tests. An EMI Control Plan, EMI/EMC Test Plan and EMC Test Report shall be prepared by the contractor and tests will be performed in accordance with MIL-STD-462 (see 6.2).



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4.4.2.4 Tempest tests. The contractor or his representative shall test the voice digitizer in accordance with the procedures of NACSEM 5100 to demonstrate compliance with the requirements of 3.8.11 thereof. The contractor shall prepare a Tempest test plan, Tempest test facility certification report, Tempest test setup ambient signal control certification report, and an EUT evaluation report. The Tempest test plan and test facility certification report shall be prepared prior to the start of Tempest testing (see 6.2).

4.4.2.5 Intelligibility test. A Diagnostic Rhyme Test shall be conducted at 2400 bps back-to-back and at $10^{-2}$ bit error rate employing a minimum of five speakers and eight listeners. The complete VODIG unit shall be operational during the test.

4.4.2.6 Loudspeaker test. Maximum intelligible volume shall be at least 76 db above $10^{-16}$ watt/cm$^2$. The maximum volume shall be measured in a sound absorbing booth or chamber. The sound level meter shall be positioned at a distance of two meters from the loudspeaker and the volume turned to maximum at time of measurement.

4.4.3 Environmental inspection. Environmental inspection shall be as listed in Table IV and shall be performed on units which have been subjected to and passed production inspection. Environmental tests shall be approved in accordance with 4.1.4.

Table IV - Environmental Inspection

| Inspection | Reference |
|---|---|
| Temperature | 4.7.2 |
| Humidity | 4.7.3 |
| Shock | 4.7 |
| Vibration | 4.7.4 |
| Intelligibility | 4.4.2.5 |

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments produced or fraction thereof for all inspections except shock and vibration. For shock and vibration, one equipment shall be selected from each successive 200 equipments produced or fraction thereof.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative the results thereof and details of the corrective action taken to correct units of production which were manufactured with the same conditions, materials, and processes. If the quality assurance representative does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

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4.4.3.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to and passed environmental tests may be accepted on the contract provided they are re-subjected to and pass production inspection. See 4.4.2.2 for rework limits.

4.4.4 Inspection for preparation for delivery. Inspections shall be conducted to ensure conformance with the requirements of Section 5.

4.5 Reliability test procedures. The reliability tests shall be conducted using the procedures, schedules, and performance limits specified herein as a minimum.

4.5.1 Failure. Failure shall be defined as any malfunction or parameter deviation which prevents the equipment from performing within the operational requirements set forth in this specification. All failures shall be considered relevant unless determined by the command or agency concerned to be caused by a condition external to the equipment under test.

4.5.2 Reliability test plan. Prior to the start of reliability testing, the contractor shall prepare a detailed reliability test plan in accordance with MIL-STD-781 (see 6.2).

Table V - Reliability Test

| Test | Specified Performance Requirements |
|---|---|
| Synchronization | 3.2.9, 3.2.14 |
| Out of synchronization detection | 3.2.16 |
| Self test | 3.8.1.1 |

Debugging and burn-in are to be performed on all production equipment prior to delivery to the Government. No preventive maintenance shall be performed during the reliability test.

4.5.3 Reliability qualification tests. Reliability qualification testing shall be a fixed time test of 500 hours accumulated simultaneously on each of the first four production units for a total test time of 2000 unit hours. One failure shall be allowed for acceptance, two or more failures shall be cause for rejection. Testing shall be performed at test level C of MIL-STD-781. Input voltage cycling is not required. The Loop-Back test shall be run continuously during the reliability test, except when the parameters of Table V are being measured. No production equipments may be shipped prior to successful completion of the qualification test. The performance parameters of Table V shall be measured once each 24 hours.



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4.5.4 Reliability sampling tests. Two units shall be randomly selected from every third month's production for testing in accordance with test level C, of MIL-STD-781. The performance parameters of Table V shall be measured once each 24 hours. The Loop Back test shall be performed continuously while the equipments are in reliability test, except when the parameters of Table V are being measured. The sampling test shall be a fixed time test of 2000 unit hours accumulated by testing the two units simultaneously for 1000 hours each. One failure shall be allowed for an accept decision, two or more failures shall be cause for rejection.

4.5.5 Reliability test reports. The contractor shall prepare test reports for reliability testing in accordance with MIL-STD-781 (see 6.2).

4.5.6 Failure analysis. All failures shall be analyzed, with the results of analysis retained by the contractor and available for Government review as required.

4.5.7 Burn-in. The contractor may elect to operate the reliability test samples for a certain time prior to the initiation of reliability testing. Any such burn-in applied to test samples must be applied to all items of equipment submitted for acceptance. Failures occurring during this period may be considered non-relevant but shall be recorded and reported. Burn-in time for all items of equipment submitted for acceptance shall be at the same levels of environment and for the same integral number of cycles as the burn-in conducted on the test samples.

4.5.8 Disposition of tested equipment. Upon successful completion of reliability tests, the equipment shall be returned to the production area for reconditioning and refurbishment as necessary. It shall then be subjected to the production inspection of 4.4.1 and shipped as part of the production quantity provided it meets all acceptance requirements of the contract. If a reject decision is reached, production shall cease and the whole lot from which the sample was drawn shall be rejected. A plan for corrective action shall be prepared for Government approval. Approved corrective action shall be implemented on all equipments delivered under the contract at the contractors expense. Production may resume upon successful demonstration by the contractor that the corrective action is adequate.

4.6 Inspection procedures.

4.6.1 Tests and examinations. The tests and examinations specified herein shall be conducted in accordance with the applicable inspection procedures of MIL-E-16400 except as hereinafter specified.

4.7 Shock test. The complete equipment shall be subjected to the Class I, Lightweight, Type A, test of MIL-S-901. The 5 foot hammer drop portion of the test is not required.

4.7.1 Acceptability. The equipment shall remain operable and shall successfully pass the operational and safety inspections of 4.4.1.



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4.7.2 Temperature tests. Temperature tests shall be conducted in accordance with MIL-E-16400, Class 4, except that non-operating temperatures shall be -50°C to 75°C.

4.7.3 Humidity tests. Humidity tests shall be in accordance with MIL-E-16400.

4.7.4 Vibration tests. Vibration tests shall be in accordance with MIL-E-16400.

4.8 Maintainability demonstration. The equipment and module maintainability requirements shall be demonstrated in accordance with Method 3 of MIL-STD-471. The maintainability demonstration shall be successfully completed prior to the delivery of any equipment. The maintainability demonstration shall be performed after completion of first article testing and prior to reliability testing.

4.8.1 Maintainability demonstration test plan. The contractor shall prepare a detailed maintainability test plan in accordance with MIL-STD-471. The maintainability test at the organizational level shall be performed by a technician with experience equivalent to that of a third class Petty Officer. The plan shall include provisions to demonstrate the ERTs for the equipment and all repairable modules. (see 6.2).

4.8.1.1 Equipment maintainability. The equipment maintainability requirements shall be demonstrated by replacement of modules and chassis-mounted electronic, electrical and electro-mechanical components and parts at the organizational level. A list of 50 faults shall be prepared for the equipment maintainability demonstration in accordance with Appendix A of MIL-STD-471. The command or agency concerned or its representative shall use this list as a guide to select a sample of 20 faults to be used in this part of the maintainability demonstration. For fault simulation, a number shall be assigned to each component and part within a module or chassis-mounted component and part. Fault simulation for modules shall be accomplished by introduction of faulty parts corresponding to the number selected. This part of the demonstration shall be conducted in a space environment (including clearance) similar to that in which the equipment will be installed.

4.8.1.2 Module maintainability. The module maintainability requirements shall be demonstrated by replacement of piece parts at the organizational, intermediate or depot level as appropriate. A list containing a minimum of 5 faults shall be prepared for each repairable module. Fault selection and simulation for each repairable module shall be in accordance with the procedure for the equipment maintainability demonstration. Using the list as a guide, a total of 20 faults shall be selected by the command or agency concerned for the module maintainability demonstration.

4.8.2 Technicians. The command or agency concerned reserves the right to provide and select the technicians to perform either or both the equipment and module maintainability demonstration. The contractor will provide adequate training to qualify the military technicians(s) to perform the maintainability demonstration.



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4.8.3 Technical documentation. Technical documentation to be used shall be limited to the equipment technical manual.

4.8.4 Rejection. If a reject decision is reached, the command or agency concerned shall be immediately notified. The contractor shall, at no additional cost to the Government:

(a) Develop a plan for design or correction of all deficiencies.
(b) Upon approval of the plan perform corrective action and repeat the demonstration until an accept decision is reached.

4.8.5 Maintainability demonstration report. A maintainability demonstration report shall be prepared in accordance with MIL-STD-471 (see 6.2). The report shall include as analysis of the statistical distribution of the repair time.



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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and Packaging, Packing, and Marking. Unless otherwise specified, preparation for delivery shall be in accordance with MIL-E-17555 and MIL-STD-794 (see 6.1).

6.0 NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

(a) Title, Number and date of this specification
(b) Number of first article samples to be submitted, if other than specified in 4.3
(c) Levels of preservation, packaging, Packing and marking required (see 5.1)

6.2 Contract data requirements. Data generated by this document are not deliverable unless specified in the contract data requirements list (DD Form, 1423) or the contract schedule. The data required by this specification include, but are not restricted to the following:

(a) Reliability program plan (see 3.4.2)
(b) Reliability prediction (see 3.4.3)
(c) Reliability reports (see 3.4.4)
(d) Maintainability program plan (see 3.5.4)
(e) Maintainability prediction (see 3.5.4.1.1)
(f) Maintainability analysis (see 3.5.4.1.2)
(g) Maintainability reports (see 3.5.4.3)
(h) Non-standard parts (see 3.5.6.1)
(i) Electron tubes (see 3.5.6.4)
(j) Microelectronic devices (see 3.5.6.7)
(k) Calibration and test procedures (see 3.8.2.1)
(l) Instrument calibration procedures (ICP) (see 3.8.2.2)
(m) LRI fault location test procedures (see 3.8.3.1)
(n) Shop LRI performance and quality test procedures (see 3.8.4.1)
(o) Shop replaceable item fault isolation and test procedures (see 3.8.5)
(p) Quality plan (see 4.1.2)
(q) Test procedures (see 4.1.4)
(r) First article inspection report (see 4.3.1)
(s) Production control inspection report (see 4.4.2.1)
(t) Electromagnetic compatibility tests (see 4.4.2.3)
(u) Tempest tests (see 4.4.2.4)
(v) Reliability test plan (see 4.5.2)



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(w) Maintainability demonstration test plan (see 4.8.1)
(x) Maintainability demonstration report (see 4.8.5)

6.3 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 Instruction for models. The first assembled VODIG shall be subjected to First Article Inspection and Acceptance Tests. The first assembled model upon satisfactory completion of the contractors first article acceptance tests shall be transported to a Navy laboratory facility for further testing. Delivery or destination will be specified in the contract or order.

6.5 Environmental inspection. Approval to ship may be witheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

6.6 Intended Use. The VODIG is intended for use both shipboard and at shore installations.

6.7 Definitions.

6.7.1 Microcircuits. A microcircuit is a small circuit having a high equivalent circuit element density, which is considered as a single part composed of interconnected elements on or within a single substrate to perform an electrical circuit function. (Typically, this includes intergrated circuits packaged in TO cans, flat packs or dual-in-line packs.)

6.7.2 Microcircuit module. A microcircuit module is an assembly of microcircuits or microcircuits and discrete conventional electron devices, performing one or more distinct functions in an electronic equipment, constructed as an independently packaged, replaceable unit. (Typically, this includes plug-in printed wiring boards and circuit card assemblies).

6.7.3 TMDE (Test, Measurement and Diagnostic Equipment) are all instruments, devices (electrical, electronic, mechanical, magnetic or chemical, etc.), tools and interface equipment, used to: test, observe, measure, monitor, modify, generate, calibrate and control physical properties, movements and other characteristics. This includes power sources, modifiers, scalers, jigs, templates, monitoring devices, waveguides, standards and programming software required to test, measure and diagnose.

6.7.4 LRI (Line Replaceable Item) is that item which has been designated to be replaced at the organizational level of maintenance.



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6.7.5 SRI (Shop Replaceable Item) is that item which has been selected to be replaced or repaired at the intermediate or depot level of maintenance.

Preparing activity:
Navy-EC

(Project 5895-N229)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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